PrepLang: a data-aware compiler for data-preparation pipelines that reuses its proofs on new batches

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Abstract

PrepLang is a compiler, written in C++17, for a small language of table and image preprocessing pipelines. It reads the input data at compile time, proves or refutes the pipeline's data contracts before any data is processed, removes the checks and operations the data makes useless, and generates a pandas and OpenCV program. The generated program keeps every removal behind a guard with the input facts it depends on, so it runs unchanged on new batches and runs a removal again only on a batch that breaks its facts. The repository also holds Lean 4 proofs for a core of the analysis, a lifter for pandas scripts, a tool that composes compiled stages, a tool that releases differentially private facts to compile against, and the complete evaluation.

Keywords: data-aware compilation, data validation, speculative optimization, deoptimization, data pipelines, differential privacy

1. Motivation and significance

Data-preparation mistakes propagate silently into models [1]. A preparation pipeline is usually a script over pandas [2] and OpenCV [3] that cannot know, until it runs, whether a column it normalizes holds numbers or whether the images it saves as PNG hold float pixels; validators such as TFDV [4], Deequ [5], Great Expectations [6] and pandera [7] check the data separately from the code that transforms it. PrepLang uses a property of batch preparation that compilers leave unused: the inputs exist when the pipeline is compiled. As F# type providers do for types [8], but for values, it reads every CSV cell, Parquet footer and image header, follows types, missing values, value ranges and image properties through the pipeline, proves or refutes each

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Nr.	Code metadata description	Please fill in this column
C1	Current code version	[v2.0.0: tag this release]
C2	Permanent link to code/repository used for this code version	`https://github.com/Nikhi00718/preLang` [make it public]
C3	Permanent link to Reproducible Capsule	None; the repository has a `Dockerfile` and a CI workflow, and `eval/paper_numbers.py` regenerates every number from the raw results
C4	Legal Code License	[an OSI-approved licence, e.g. MIT, in a LICENSE file]
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	C++17 (no external libraries); Python 3 with pandas, NumPy, PyArrow, OpenCV; Lean 4; CMake
C7	Compilation requirements, operating environments and dependencies	CMake 3.16+, a C++17 compiler (tested: Visual Studio 2022 on Windows 11, GCC 15.2 on Ubuntu); Python 3.12+ with pandas (2.0+, 3.0 tested), PyArrow, OpenCV; Lean 4.34.1 (optional, for the proofs)
C8	If available, link to developer documentation/manual	`README.md` (user manual), `ARCHITECTURE.md`, `CONTRIBUTING.md`; the READMEs of `eval/`, `paper/`, `proofs/`, `docs/`
C9	Support email for questions	atnikhil06@gmail.com

Table 1: Code metadata (mandatory)

contract before any data is processed, and removes the checks and operations the data makes useless.

Such proofs hold for one file, while the pipeline runs again next month on a file with the same layout and other values. PrepLang therefore computes, for every removal, the facts about the input it depends on (its *item certificate*), and the generated program checks them on each new batch: a removal whose facts hold stays removed, one whose facts a batch breaks runs again (*selective deoptimization*, from dynamic compilers [9, 10]), and a batch is refused only for facts no run-time code can stand in for, such as a changed column type. A user writes a `.prep` program, compiles it against the current batch, reads the diagnostics and deploys the generated program for every later batch. The theory (with its Lean proofs) and the full evaluation are documented in the repository (`paper/`, `docs/`); this article describes the software.

2. Software description

2.1. Software architecture

The compiler (`src/`, about 11,800 lines of C++17 without external libraries) runs lexer, LL(1) parser, semantic analysis, IR, optimizer, certifier and code generator in turn. The *schema* readers scan CSV files in parallel and read Parquet footers and PNG and JPEG headers; the *semantic analyzer* computes the abstract dataset state (types, missing values, value intervals, image properties) and decides every contract; the *optimizer* removes, speculatively, what the state shows to have no effect; the *certifier* re-runs the analysis

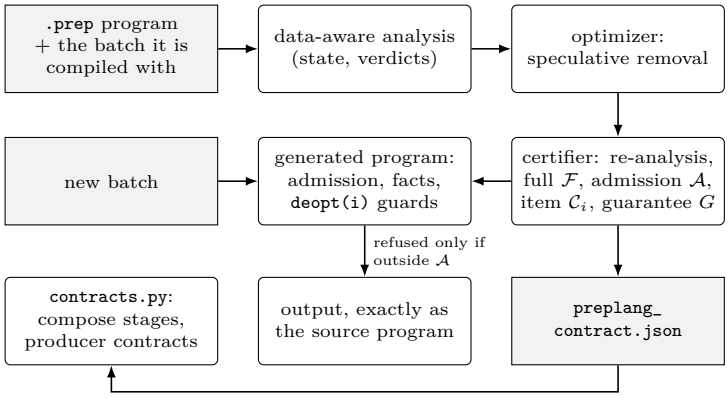


Figure 1: How PrepLang compiles once and reuses the result. The certifier re-runs the analysis with weaker input facts, reading no data; on a new batch, the generated program runs a removed instruction i only if the batch breaks its item certificate $\mathcal{C}_i$.

under weakened input facts, reading no data, to find the admission certificate and one item certificate per removal (Fig. 1); the *back end* generates one Python program that admits a batch and guards every removed instruction with `deopt(i)`, and writes the certificates to `preplang_contract.json`.

The repository also holds `proofs/` (1477 lines of Lean 4 [11]: selective deoptimization is exact for any monotone and sound analysis, and the analysis of a core of the language is sound and monotone), `tests/` (297 checks in 10 suites, with randomized soundness tests), a lifter of pandas scripts (`frontend/`), the composition and release tools (`tools/`), a local editor app (`studio/`) and the evaluation (`eval/`).

2.2. Software functionalities

- **Compile-time checking:** misspelled or removed columns, type errors and broken contracts (`NOT_NULL`, `BETWEEN`, `UNIQUE`, `REFERENCES`, schemas, image channels and sizes, match rates of images and labels), lookup joins on keys of other types or with repeated values, reported with the line, the offending row and a hint; groups (`AGGREGATE`) keep their facts.
- **Data-driven optimization:** proven checks and no-effect operations are removed, unused columns are not loaded, image pipelines are streamed; rewrites that change the output need `-approx`.
- **Reuse:** admission by certificate, and re-enabled removals with the reason printed.
- **Composition:** `tools/contracts.py` checks one stage's published guarantees against the next stage's certificates and writes the contract a producer must meet.

- **Private compilation:** the data owner releases differentially private facts at the cells the program's numbers define (`-queries`, `tools/release_facts.py`); the analyst compiles with `-facts` without opening any input file, and every run stays exact.
- **Inspection:** every phase can be printed (`-tokens` to `-python`), with an execution plan (`-explain`) and the files a run would touch (`-dry-run`).

3. Illustrative example

```
LOAD "students.csv" AS students;
EXPECT students.age NUMBER;
EXPECT students.marks BETWEEN 0 AND 100;
EXPECT students.name NOT_NULL;
FILTER students WHERE marks >= 50;
EXPECT students.marks BETWEEN 60 AND 100;
SAVE students AS "clean.csv";
```

Compiled against an eight-row `students.csv` (from `examples/tabular/`), PrepLang proves the first three contracts and checks the fourth at run time, since after the filter it only knows that the marks lie in $[50, 95]$. It removes the range and not-null checks speculatively, and `-certificate` shows the facts the range check needs: marks at least 0 and at most 100. Run on a later batch in which a mark is 104, the generated program re-enables the check and stops where a fresh compilation would:

```
Deoptimized line 3: EXPECT students.marks BETWEEN 0 AND 100:
  marks goes up to 104, above 100
ContractError: line 3: students.marks has values outside 0 to 100
```

A batch within the facts runs with both checks removed; a batch whose `age` holds text is refused, since no generated code can stand in for the analysis of another type.

4. Impact

The evaluation in `eval/` answers six questions (`eval/evaluate.py` writes the answers with every number below):

- **Correctness: is the output right?** 2,000 random programs (with joins and groups), compiled with and without the optimizer, agree cell by cell with a reference interpreter that shares no code with the compiler, and reused on a second batch none of the 1,535 runs differs from it; before two fixes this test led to (a pruning bug, a pandas 3 incompatibility), 109 optimized runs were wrong. Reused on new batches, 64,179

runs after 4,879 private compilations and every run of the experiments below did what a fresh compilation does.

- **Performance.** An optimized table pipeline runs in 0.47 s instead of 0.79 s after Python start-up, and an image pipeline as fast as a hand-written streaming script, in 10 MiB instead of the 922 MiB of a script that loads every image.
- **Optimization effectiveness.** On 435 examples and lifted programs the optimizer removes 14% of the instructions, 73% of the run-time checks and 2% of the columns read, with identical output.
- **Practical usefulness.** 18 of 44 real failures from GitHub issues are reported before any data is processed; lifted pandas scripts from GitHub reproduce their results on new batches in 3198 of 3212 runs, and in all 3255 monthly runs of 72 scripts written by others; on 16 million Parquet rows a misspelled column is reported in 0.02 s instead of after 2.81 s.
- **Comparison.** On 238 real monthly batches the compiled programs handle 221 (93%) as a fresh compilation would, with 0 wrong; Great Expectations and pandera refuse 92–93% of the months a fresh compilation runs, and TFDV admits months on which the program then differs. All 24 injected mistakes are reported at compile time, where a script-style program crashes on 20 and writes wrong output on 4. Compiling from private facts, the release at the program's numbers keeps every removal of the taxi and traffic pipelines at $\varepsilon = 1$, where fixed bins keep 60% and 83%; HDMM [12] and AIM [13], given the same cells, do no better.
- **Overhead.** Compiling takes a median of 30 ms and at most 8 MiB (65 MiB for a million CSV rows, 6 MiB for a 346 MiB Parquet file, of which only the footer is read); admitting a new batch changes run time by −1%.

One script, `eval/paper_numbers.py`, regenerates every number from the raw results, with the baselines in pinned environments (`eval/README.md`). PrepLang is a research prototype, built by the author and partly within a compiler-design course; it has no users outside the author's experiments yet. Others can extend its abstract state, certifier and code generator, and compare on the same batches.

5. Limitations and conclusions

Joins are lookups on a key with one name in both tables (no many-to-many joins), groups have five aggregates, and only Python is generated; a

changed column type is refused rather than handled; the mechanized proofs cover a core of the analysis without joins and groups, and the rest is tested; the compiler is tested on Windows 11 (MSVC) and Ubuntu (GCC 15.2). Planned work is general joins, generated code that adapts to type changes, and packaging. PrepLang compiles a pipeline once and runs it on later batches exactly as a fresh compilation would, re-enabling only the removals a batch makes unsafe.

Declaration of competing interest

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